

SIMPLIFIED FLUID TRANSMISSION LINE MODEL FOR PNEUMATIC CONTROL APPLICATIONS

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ABSTRACT

The neglect of transmission lines in pneumatic controller synthesis limits the possible control performance or the length of the connecting tubes. In literature, complex and accurate models have been derived for the description of flow phenomena in tubes, but low-order models for controller design and stability analysis are rare. This paper proposes a second-order modal approximation of the one-dimensional linear resistance compressible model and includes non-linear tube friction. The model was validated in time domain as well as in frequency domain. An application for the model in pressure control was shown and compared to the state of the art.

KEYWORDS: Modeling, fluid transmission line, pneumatics, model reduction, frequency response validation

1. INTRODUCTION

In pneumatics, the influence of tubes on pressure and mass flow is a well known research field. Numerous models with various levels of detail have been derived and experimentally validated. But when it comes down to controller synthesis and stability analysis of closed-loop control (e.g. pressure control or servopneumatic positioning) the influence of transmission lines is mainly neglected or oversimplified.

An overview of different fluid transmission line models is given in [1]. The presented models vary in the assumptions made and therefore in complexity of their analytical solutions. These models give good insight into different flow phenomena but are mainly not applicable in simulation of fluidic systems. For this reason, many different approximation methods exist for the fluid transmission line models in literature. [2] reviewed four different methods. Two of them are the finite difference approximation and the lumping-by-length method. Both are afflicted with frequency errors that increase with the number of nodes respectively lumps. Further methods, i.e. the method of characteristics and a modal approximation, yielded considerably better results. The modal approximation is superior in terms of stability and ease of use.

Similar approaches were investigated by different authors. [3] chose a decoupled representation (stiffness matrix) of the line model for a modal approximation. Compared to Hullender's approximation [4] of the transfer matrix representation, they found their method to yield several simplifications while maintaining the same accuracy. The presented model is of order 17. An approximation of significantly lower order is given in [5]. The

combination of a time delay with transfer function elements for damping achieved good results that are comparable with the ones in [3]. The stiffness matrix representation was the basis in [6], too. The approximation is carried out with a variational Ritz method, reproduces the modes accurately yet leads to steady-state errors. The minimum system order is two but increases with the correction of the steady-state error. An expansion to non-linear friction was shown.

These research topics cover mainly the general behavior of pneumatic or hydraulic transmission lines for time domain simulations. [7, 8] derived an approximation of the line model with special focus on use in controller design. It was modeled as a time delay including an attenuation of the mass flow's amplitude. Laminar as well as turbulent flow was considered. It was shown that the consideration of the tube model in controller design improved the performance at the expense of high computational effort.

The goal of this paper is to provide a low-order pneumatic transmission line model that reproduces the dynamics of pressure and mass flow in time as well as in frequency domain. The model's structure should enable stability analysis and time domain controller design for pneumatic applications, i.e. servopneumatic positioning or pressure regulation. The considered tube lengths vary from one to five meters and are typical for this application field. The basis for the simplified transmission line model is the one-dimensional linear resistance compressible model [1]. Although this model is of lesser overall accuracy than models with frequency dependent friction, it fits the purpose of a low-order approximation. Subsequently, the tube dynamics are extended by a non-linear friction model.

2. FLUID TRANSMISSION LINE MODEL

The one-dimensional linear resistance compressible model is given by the first-order partial differential equation (PDE) system

$$\frac{\partial p}{\partial x} = -\frac{1}{A} \frac{\partial \dot{m}}{\partial t} - R \dot{m}, \quad (1a)$$

$$\frac{\partial \dot{m}}{\partial x} = -\frac{A}{c^2} \frac{\partial p}{\partial t}, \quad (1b)$$

where $p = p(x, t)$ and $\dot{m} = \dot{m}(x, t)$ are pressure and mass flow, dependent on time t and the paraxial coordinate x , A the tube's inner cross-sectional area and c the sonic speed. The coordinate $x \in [0, l]$ is bounded by the length l of the tube. These variables are designated in the schematic drawing of the tube in Figure 1. The line

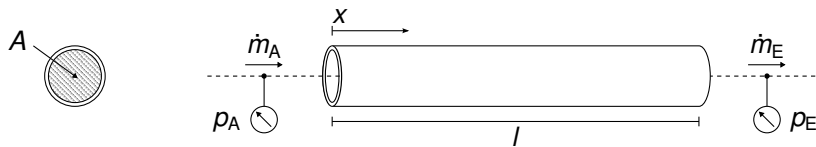


Figure 1. Designation of pressures, mass flows and geometric parameters of a tube.

resistance R is constant hence flow is assumed to be laminar. This assumption is necessary for the existence of the algebraic solution that, evaluated at the border conditions

$$p_A = p(0, t), \quad (2a)$$

$$p_E = p(l, t), \quad (2b)$$

$$\dot{m}_A = \dot{m}(0, t), \quad (2c)$$

$$\dot{m}_E = \dot{m}(l, t), \quad (2d)$$

is stated as follows

$$p_E = \frac{1}{\cosh(\Gamma(s)l)} \left(p_A - \dot{m}_E Z(s) \sinh(\Gamma(s)l) \right), \quad (3a)$$

$$\dot{m}_A = \frac{1}{\cosh(\Gamma(s)l)} \left(\dot{m}_E + p_A \frac{1}{Z(s)} \sinh(\Gamma(s)l) \right). \quad (3b)$$

The mixed boundary conditions are needed for simulation in order to obtain a causal system behavior. The dimensionless propagation function $\Gamma(s)$ and the characteristic impedance function $Z(s)$ characterize the connection between pressure and mass flow in the tube and are given by

$$\Gamma(s) = \frac{1}{c} \sqrt{(AR + s)s}, \quad (4a)$$

$$Z(s) = \frac{c}{A} \sqrt{\frac{AR + s}{s}}. \quad (4b)$$

The algebraic solution is a transcendent two-by-two transfer matrix with infinitely many poles and eigenfrequencies. From a control theory point of view the location of the poles is of great significance. By solving the equation

$$\cosh(\Gamma(s)l) = 0, \quad (5)$$

one can calculate the transfer function's poles respectively the natural frequency and damping of the resonance i by

$$\omega_{0,i} = \frac{\pi c (2i + 1)}{2l}, \quad (6a)$$

$$D_i = \frac{R/A}{c \pi (2i + 1)}. \quad (6b)$$

In the following, the exact solution (3) is taken as a reference for all approximations.

2.1. Modal approximation

While the transmission line model (3) gives a good insight into the fluid dynamics in the tube, it is not suitable for time domain stability analysis and controller synthesis. Such tasks require a low-order system that reproduces the general behavior and matches the dominant eigenvalues of the exact solution.

The modal approximation was detailed in [9] for the inviscid line model. It is based on a truncated Taylor series (TTS) of the hyperbolic functions. The result did not produce a good match for the first resonance frequency. In the following, the three different approximations in Table 1 of the hyperbolic functions are applied and analyzed. Approximation 1 is the truncated Taylor series. It is added for completion and used for comparison with the other methods. The second approximation uses the truncated product series (TPS) for the hyperbolic cosine

Table 1. Different approximations of the hyperbolic sine and cosine.

Function	Product series	Taylor series	Appr. 1 (TTS)	Appr. 2 (TPS)	Appr. 3 (LTTS)
$\cosh(x)$	$\prod_{k=1}^{\infty} \left(1 + \frac{4x^2}{(2k-1)^2 \pi^2} \right)$	$\sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}$	$1 + \frac{x^2}{2}$	$1 + \frac{4}{\pi^2} x^2$	1
$\sinh(x)$	$x \prod_{k=1}^{\infty} \left(1 + \frac{x^2}{(k\pi)^2} \right)$	$\sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$	x	x	x

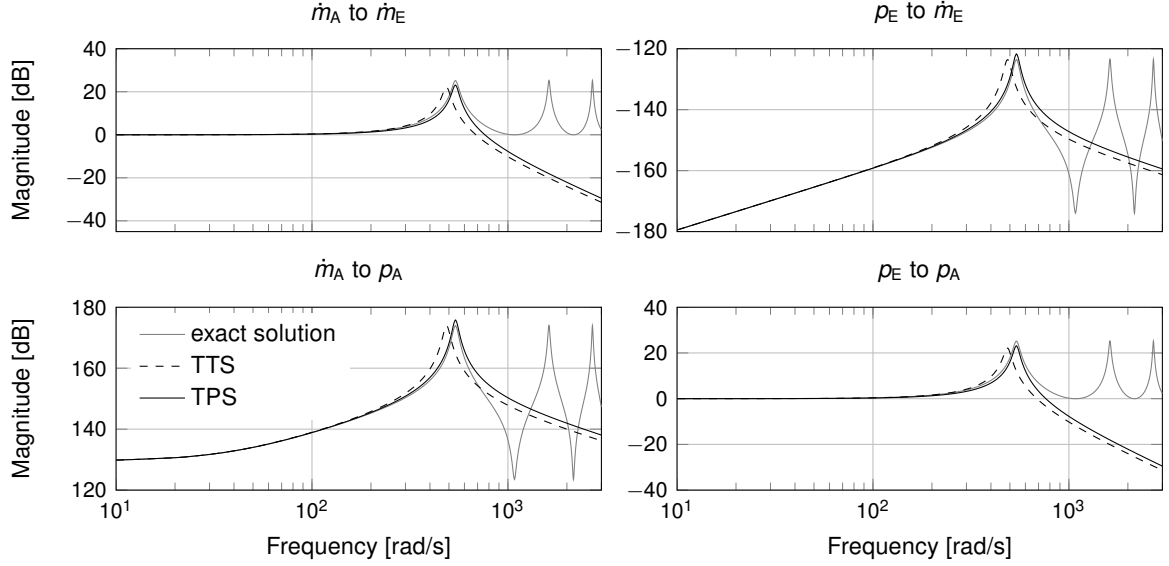


Figure 2. Magnitude plot of the exact solution, the TTS model and the TPS model. Tube of length 1 m and inner diameter 4 mm.

and the Taylor series for the hyperbolic sine. It results in a transfer matrix

$$\begin{pmatrix} \dot{m}_E \\ p_A \end{pmatrix} = \begin{pmatrix} \frac{\pi^2 c^2}{4 I^2 s^2 + 4 A R I^2 s + \pi^2 c^2} & -\frac{\pi^2 A I s}{4 I^2 s^2 + 4 A R I^2 s + \pi^2 c^2} \\ \frac{\pi^2 c^2 I (A R + s)}{A (4 I^2 s^2 + 4 A R I^2 s + \pi^2 c^2)} & \frac{\pi^2 c^2}{4 I^2 s^2 + 4 A R I^2 s + \pi^2 c^2} \end{pmatrix} \begin{pmatrix} \dot{m}_A \\ p_E \end{pmatrix}, \quad (7)$$

consisting of four second-order systems. The natural frequency and damping of the poles

$$\omega_0 = \frac{\pi c}{2 I}, \quad (8a)$$

$$D = \frac{R I A}{c \pi} \quad (8b)$$

match those of the exact solution's first resonance.

Figure 2 shows the frequency response of the exact solution, the TPS model (7) and the TTS model using the truncated Taylor series for both hyperbolic functions. The accuracy of the model was increased with the the product series expansion. Only small deviations concerning the amplitude of the resonance frequency are left. In contrast to the TTS approximation in [9], the TPS method results in a better fit to the first resonance frequency. The algebraic controllability and observability analysis shows that the minimal state space realization of (7) is of order four.

The third approximation (LTTS) substitutes the hyperbolic cosine by the constant 1 and is therefore the lowest-order approximation. The application to (3) and transformation into time domain yields the second-order system

$$\frac{d\dot{m}_A}{dt} = \frac{A}{I} (p_A - p_E - R I \dot{m}_A), \quad (9a)$$

$$\frac{dp_E}{dt} = \frac{c^2}{A I} (\dot{m}_A - \dot{m}_E). \quad (9b)$$

The result of this approximation is closely related to the finite volume method [10] which divides the transmission line alongside the x coordinate into N parts. Each part describes a finite volume with pressure and mass flow dynamics. With increasing N , the accuracy of the solution increases as well as the order of the resulting system. For $N < 5$ the error of the first resonance frequency is greater than 10 %. Setting the number of finite volumes to $N = 1$, the finite volume method produces the same model as the introduced LTTS method.

In contrast to the previous transfer function, the boundary conditions change from $\dot{m}_A$ and p_E to $\dot{m}_E$ and p_A . To

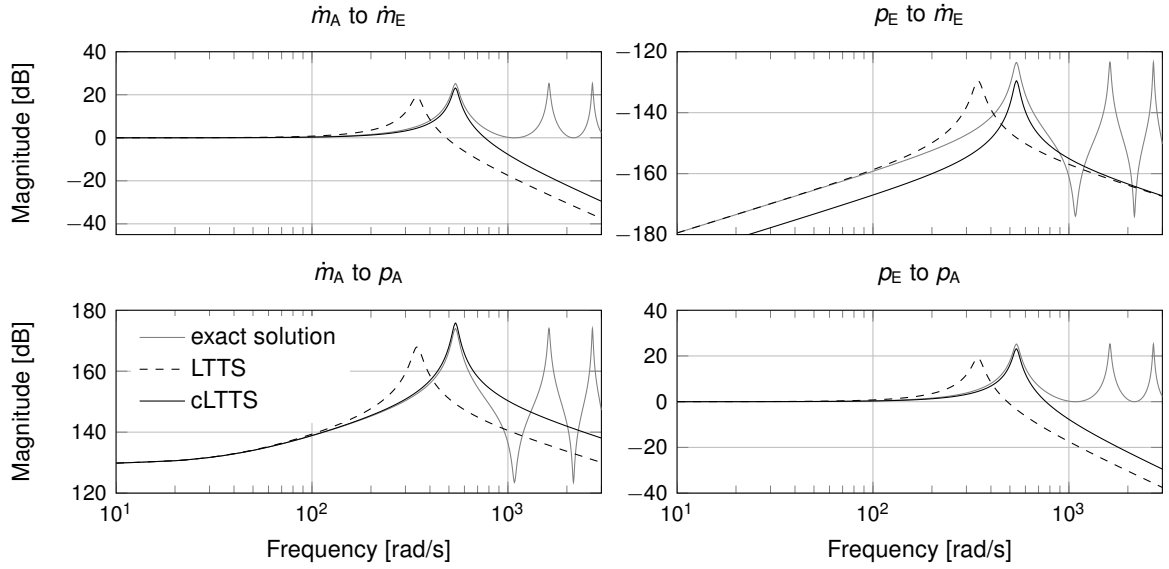


Figure 3. Magnitude plot of the exact solution, the LTTS model and the cLTTS model. Tube of length 1 m and inner diameter 4 mm.

get the same boundary conditions as in (7), model (3) needs to be solved for $\dot{m}_A$ and p_E and approximated, achieving the corrected LTTS (cLTTS) model

$$\frac{d\dot{m}_E}{dt} = \frac{A}{I} (p_A - p_E - R I \dot{m}_E), \quad (10a)$$

$$\frac{dp_A}{dt} = \frac{k c^2}{A I} (\dot{m}_A - \dot{m}_E). \quad (10b)$$

Since the low-order approximation leads to an error concerning the resonance frequency, the factor k was inserted supplementary to correct that error. The natural frequency and damping of the eigenvalues are given in dependence of k as

$$\omega_0 = \sqrt{k} \frac{c}{I}, \quad (11a)$$

$$D = \frac{R I A}{2 c \sqrt{k}}. \quad (11b)$$

By comparing Equations (6) and (11), one can calculate

$$k = \frac{\pi^2}{4}, \quad (12)$$

such that the eigenvalues of the approximation fit the exact solution. This factor already showed up in the truncated product series expansion and leads to the improved matching of the cLTTS model. The transformation of (10) into frequency domain makes the difference to the TPS model clear. The transfer matrix

$$\begin{pmatrix} \dot{m}_E \\ p_A \end{pmatrix} = \begin{pmatrix} \frac{\pi^2 c^2}{4 I^2 s^2 + 4 A R I^2 s + \pi^2 c^2} & -\frac{4 A I s}{4 I^2 s^2 + 4 A R I^2 s + \pi^2 c^2} \\ \frac{\pi^2 c^2 I (A R + s)}{A (4 I^2 s^2 + 4 A R I^2 s + \pi^2 c^2)} & \frac{\pi^2 c^2}{4 I^2 s^2 + 4 A R I^2 s + \pi^2 c^2} \end{pmatrix} \begin{pmatrix} \dot{m}_A \\ p_E \end{pmatrix} \quad (13)$$

equals (7) except for the gain of the upper right element.

Figure 3 shows the magnitude plot of the cLTTS model (13) and LTTS model against the exact solution's magnitude plot (3). The uncorrected solution differs in terms of resonance frequency. The correction leads to a far better approximation in three transfer functions and only results in a magnitude error of the transfer function from pressure to the mass flow at the end of the tube. This transfer function will later be eliminated by the coupling of the tube with a terminal volume.

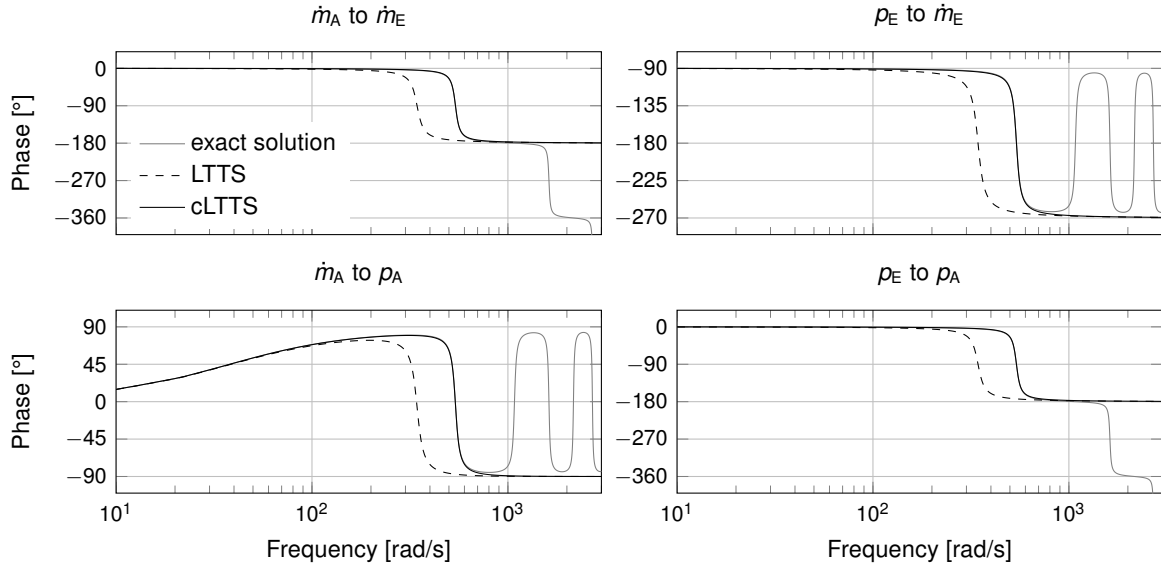


Figure 4. Phase plot of the exact solution, the LTTS model and the cLTTS model. Tube of length 1 m and inner diameter 4 mm.

In Figure 4, the phase plot of the exact solution, the LTTS and the cLTTS model is shown. The corrected approximation fits the phase response of the exact solution precisely up to the first resonance frequency. For higher frequencies, the phase response of the approximations stays on the same level while the exact solutions phase response oscillates or decreases respectively. In contrast to the TPS model, the cLTTS model can be realized with only two first-order differential equations. It is remarkable that the second-order model has the same transfer function structure as the fourth-order model used in literature.

2.2. Non-linear friction modeling

Until now, both models only consider friction in form of linear damping that depends on mass flow. This is valid for laminar flow. In pneumatic applications like pressure control or servopneumatics, high flow rates over a short period of time are more common. For this reason, the model must be valid for turbulent flow, too. There are two common methods that describe the pressure drop in tubes, considering laminar as well as turbulent flow — the Hagen-Poiseuille respectively the Darcy-Weisbach equation (λ method) and the pneumatic related {C,b} method. The λ method was applied in [10] for a high-order model of a pneumatic transmission line. Dependent on the Reynolds number, different linear and non-linear equations must be evaluated for the calculation of the pressure drop.

A more compact method for obtaining the pressure drop is the modeling of the tube as a pneumatic nozzle. The mass flow

$$\dot{m} = C \rho \Psi(q, b) p_{\text{high}} \quad (14)$$

through the nozzle depends on the sonic conductance C , the critical pressure ratio b , the pressures p_{low} and p_{high} respectively their ratio $q = p_{\text{low}}/p_{\text{high}}$. An empirical formula for sonic conductance and critical pressure ratio dependent on tube geometry is given in [11] as

$$C = \frac{0.029 d^2}{\sqrt{\frac{l}{d^{5/4}} + 510}}, \quad (15a)$$

$$b = \frac{474 C}{d^2}. \quad (15b)$$

To calculate the pressure drop via the {C,b} method, Equation (14) needs to be solved for the pressure. Since

the flow function

$$\Psi(q, b) = \begin{cases} 1 & \text{if } q < b \\ \sqrt{1 - \left(\frac{q-b}{1-b}\right)^2} & \text{if } q \geq b \end{cases} \quad (16)$$

is not invertible for pressure ratios smaller than the critical pressure ratio, a polynomial approximation

$$\tilde{\Psi}(p_{\text{high}}, p_{\text{low}}) = \frac{a_{\Psi} (p_{\text{high}} - p_{\text{low}})}{c_{\Psi} p_{\text{high}} - p_{\text{low}}} \quad (17)$$

is used for the inversion. The factors a_{Ψ} and c_{Ψ} must be chosen appropriately. The resulting pressure drop is expressed by

$$\Delta p = \Phi(\dot{m}_E, p_E). \quad (18)$$

In contrast to the λ method which depends solely on the mass flow, the {C,b} method depends on mass flow and pressure. Combining the cLTTS model (10) and the pressure drop (18) adds up to the non-linear second-order model

$$\frac{d\dot{m}_E}{dt} = \frac{A}{I} (p_A - p_E - \Phi(\dot{m}_E, p_E)), \quad (19a)$$

$$\frac{dp_A}{dt} = \frac{\pi^2 c^2}{4 A I} (\dot{m}_A - \dot{m}_E). \quad (19b)$$

2.3. Line model with terminal volume

For experimental validation or pneumatic applications, the tube model cannot be considered without the influence of a terminal volume. Coupling mass flow and pressure at the end of the tube with a simple model of a pneumatic volume

$$\frac{dp_E}{dt} = \frac{n R_a T_0}{V_V} \dot{m}_E, \quad (20)$$

leads to the three transfer functions

$$G_{P_A}(s) = P_A(s) / \dot{M}_A(s), \quad (21a)$$

$$G_{P_E}(s) = P_E(s) / \dot{M}_A(s), \quad (21b)$$

$$G_{\dot{M}_E}(s) = \dot{M}_E(s) / \dot{M}_A(s). \quad (21c)$$

Figure 5 shows the frequency response of $G_{P_A}(s)$ with different terminal volume sizes. The resonance frequencies increase with decreasing volume size. In comparison to Figure 3, an additional antiresonance can be seen. Since the same characteristic can be observed in measurements as well as in the model, this effect is not a model artifact but real system behavior. The effect gets significant as soon as the terminal volume ap-

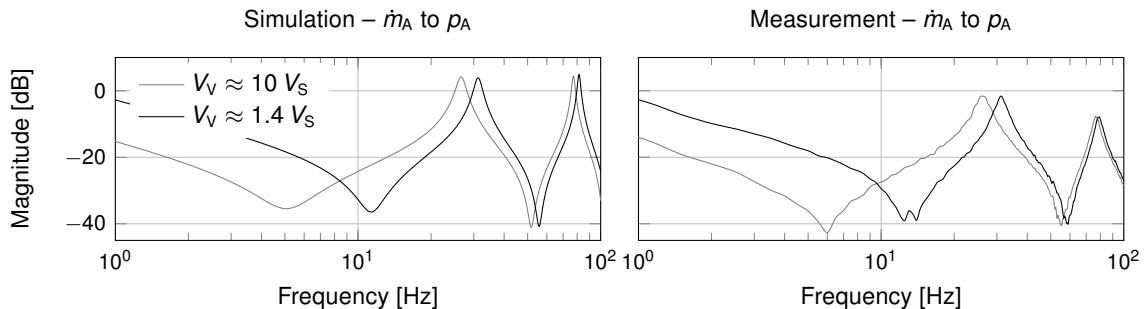


Figure 5. Magnitude plot of $G_{P_A}(s)$ of a tube (inner volume $V_S \approx 0.07$ L) with a big ($V_V = 0.75$ L) and a small ($V_V = 0.1$ L) terminal volume. Simulation on the left and measurement on the right side.

proaches the tube's inner volume. With an infinitely big volume, the resonance frequencies converge towards the resonance frequencies of the tube model without volume.

3. EXPERIMENTAL VALIDATION OF THE TUBE MODEL

The experimental validation is based on time and frequency domain responses. For the frequency response identification the valve slide is excited by band limited white noise with a maximum frequency of 100Hz. The amplitude of the excitation is chosen low enough for laminar flow conditions to apply and non-linear friction effects to be neglected. Time domain is validated with a sinusoidal excitation of the valve slide. Those measurements are used to identify the non-linear friction effects resulting in a pressure drop. The validated transfer functions in the following section are $G_{P_A}(s)$ from valve mass flow $\dot{m}_A$ to valve pressure p_A and $G_{P_E}(s)$ from valve mass flow $\dot{m}_A$ to terminal volume pressure p_E . The transfer function of the mass flows $\dot{m}_A$ and $\dot{m}_E$ is neglected since the reactionless determination of the mass flow $\dot{m}_E$ is not possible in this test setup. In frequency domain, pressure and mass flow are normalized to 1×10^{-5} Pa and 1×10^3 kg/s.

The test rig, shown in Figure 6, consists of a proportional valve, the pneumatic tube and a terminal volume. The

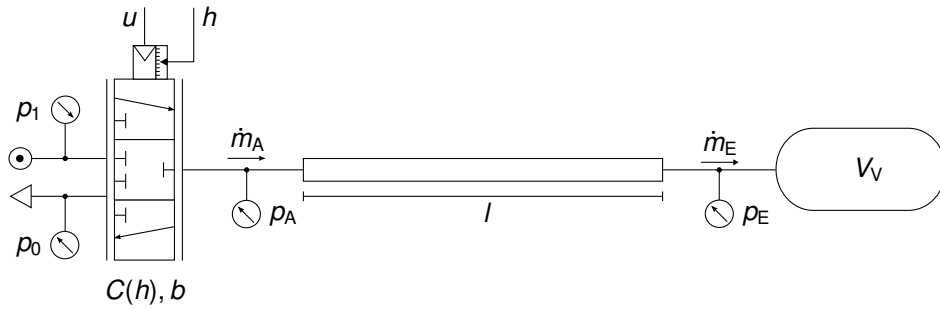


Figure 6. Schematic representation of the test rig.

mass flow $\dot{m}_A$ through the valve is calculated with Equations (14) and (16) with the measurement of the valve slide h , the supply pressure p_1 , the exhaust pressure p_0 and the pressure p_A . In contrast to direct measurement of the mass flow with a sensor, the calculated quantity has the higher dynamics that are needed for frequency response determination.

Figure 7 shows the model's and the measured frequency responses of $G_{P_A}(s)$ and $G_{P_E}(s)$. The tube has a length of 3 m and an inner diameter of 5.7 mm. The terminal volume's size is 0.75 L. The model matches the resonance frequency of about 25 Hz and the frequency characteristic up to 50 Hz sufficiently. The position

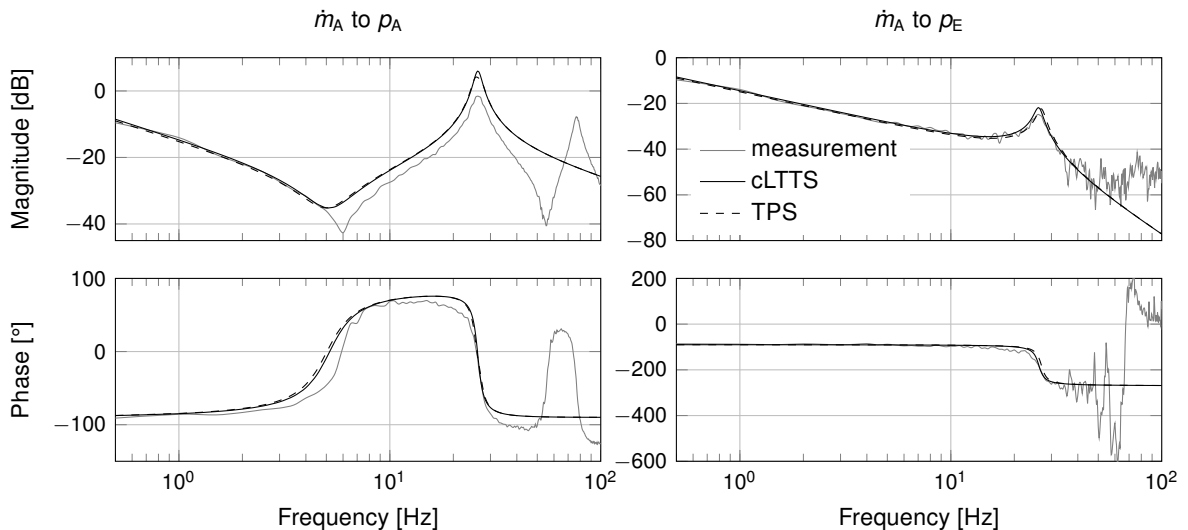


Figure 7. Frequency responses of $G_{P_A}(s)$ on the left and $G_{P_E}(s)$ on the right. Tube of length 3 m and inner diameter 5.7 mm with a terminal volume of 0.75 L.

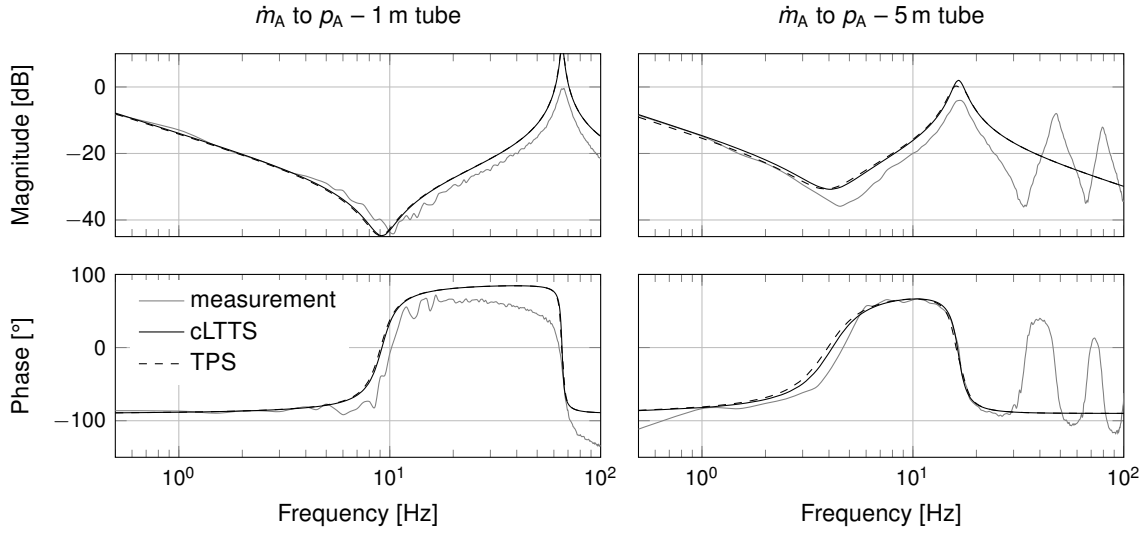


Figure 8. Frequency responses of $G_{P_A}(s)$ of a tube of length 1 m on the left and 5 m on the right. The tube's inner diameter is 5.7 mm and the terminal volume size 0.75 L.

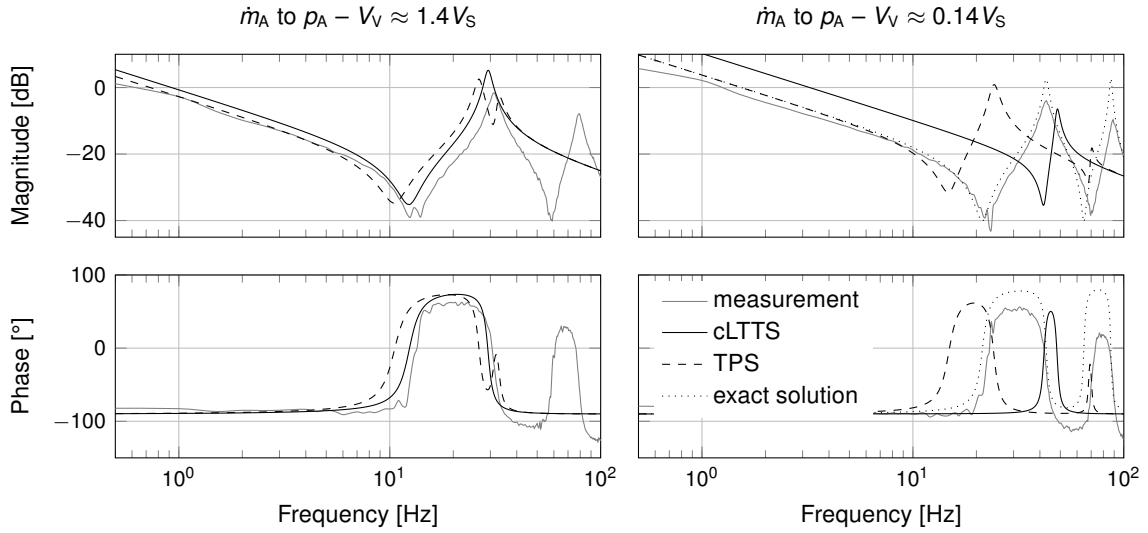


Figure 9. Frequency response of $G_{P_A}(s)$ of a tube of 3 m length and 5.7 mm inner diameter. The terminal volume size is 0.1 L on the left and 0.01 L on the right.

of the first antiresonance of $G_{P_A}(s)$ as well as the steady state amplification of $G_{P_E}(s)$ depend mainly on the terminal volume. The erroneous frequency of the antiresonance cannot be corrected without changing the low-frequency amplification of $G_{P_E}(s)$. Aside from small differences, the second-order cLTTS model and the fourth-order TPS model show the same behavior. The same is valid for the frequency response in Figure 8. Here, the measurement of two tubes with the lengths 1 m and 5 m is shown. Again, both approximations provide comparable results.

Figure 9 shows the frequency response of the same tube with two different terminal volume sizes, respectively 0.1 L and 0.01 L. The size of the first terminal volume is similar to the tube's inner volume of approximately 0.07 L. The second terminal volume is much smaller. Such ratios are an exception in pneumatic applications but they illustrate the difference between the TPS and the cLTTS model. The amplitude response of the higher-order TPS model forms two resonances: one that depends only on pipe geometry and one that also includes the terminal volume's parameters. With decreasing volume sizes, both resonance frequencies diverge from each other (Figure 9 right side). The amplitude for low frequencies matches the measurement.

While the cLTTS model replicates the frequency response on the left reasonably well, the antiresonance frequency deviates significantly on the right. Compared to the fourth-order TPS model, the resonance frequency provides a better fit. The error for low frequencies comes from the resonance correction (12). The correction resulted in an amplitude error of the transfer function from p_E to $\dot{m}_E$ that was insignificant for large terminal volume sizes. The exact solution of the transmission line model however indicates that amplitude and resonance

errors originate in the approximation. Higher-order approximations with the finite volume method will result in a better replication of the frequency characteristic for very small terminal volumes.

The time domain validation of the non-linear cLTTS model is displayed in Figure 10. The valve slide is excited with a 2 Hz sinusoid of decreasing amplitude. The measurement clearly shows the necessity of considering tube dynamics and friction. With a tube of 3 m length and 5.7 mm inner diameter, the difference in pressure at valve and terminal volume (0.75 L) is up to 400 kPa. The detail view on the right side shows that the tube model reproduces the large-signal behavior as well as small oscillations while the mass flow direction changes. The measured and simulated mass flows are depicted in the middle part. The mass flow $\dot{m}_{E,s}$ at the end of the tube is lagged and lower in amplitude than the mass flow at the beginning. Additional oscillations, similar to those of p_A , can be seen. The lower part of Figure 10 shows a comparison between the linear and the non-linear friction model. This comparison stresses the necessity of the non-linear extension, which attenuates the pressure oscillations to a large extent and results in a pressure drop of up to 2×10^5 Pa. In this example, laminar flow conditions only apply at mass flows lower than approximately 2.2 kg/s. In time domain, the neglect of turbulent flow conditions has a much larger impact than the choice of model (e.g. linear resistance, viscous compressibility, thermal effects) or the approximation method.

The validation in frequency domain showed a good agreement of the TPS and the cLTTS model with measurements. Tube lengths of 1 m, 3 m and 5 m, as well as different terminal volume sizes were considered. Only very small terminal volume sizes revealed the difference between the two models. The observed model deviations are not a big limitation, since such tube volume to terminal volume ratios are rarely found in pneumatic applications. The extension with the non-linear friction model made the tube model also valid for turbulent flow conditions and resulted in an excellent match in small- and large-signal behavior.

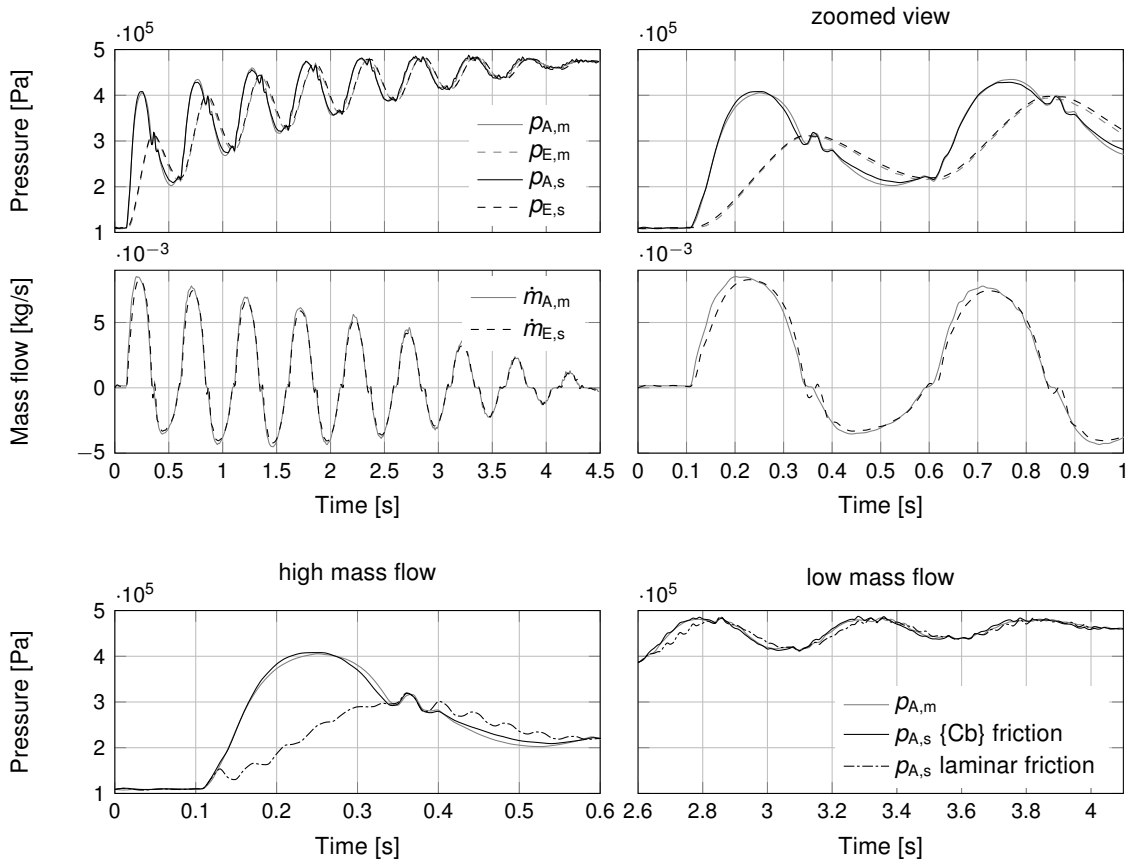


Figure 10. Validation of the pressures and mass flows at beginning and end of a tube of 3 m length and 5.7 mm inner diameter. Index m for measured and s for simulated.

4. APPLICATION IN PRESSURE CONTROL

The standard model for pressure control applications does not include tube dynamics and hence the pressures p_A and p_E as well as the mass flows $\dot{m}_A$ and $\dot{m}_E$ are considered equal. The model is given by

$$\frac{P_E(s)}{\dot{M}_V(s)} = \frac{1}{s} \frac{n R_a T_0}{V_V} \frac{1}{T_V s + 1} = \frac{k_V}{s(T_V s + 1)}. \quad (22)$$

It consists of a pneumatic volume (20) and valve dynamics, modeled as a first-order lag with time constant T_V . The input is the mass flow $\dot{m}_V$ in front of the valve and the output the pressure p_E in the volume. In reality the mass flow depends on the valve pressures and the valve slide position and is not the actual control input. This dependence can be inverted and the mass flow becomes a virtual input of the plant. With a proportional controller, that regulates the pressure in the volume, the closed-loop's transfer function is

$$G_{cl} = \frac{k_P k_V T_V^{-1}}{s^2 + T_V^{-1} s + k_P k_V T_V^{-1}} \quad (23)$$

and the controller gain can be expressed in dependence of a desired damping factor D by

$$k_P = \frac{1}{4 D^2 k_V T_V}. \quad (24)$$

The derived control law is now identically applied to the nominal model (22) and to a model that is extended by the linear tube dynamics (10). The tube has a length of 2 m and an inner diameter of 5.7 mm. Figure 11 shows on the left the desired pressure trajectory $p_{E,d}$ and the time responses of the control loop with and without tube dynamics. The pressure transient of the nominal model shows the expected behavior of a second-order lag

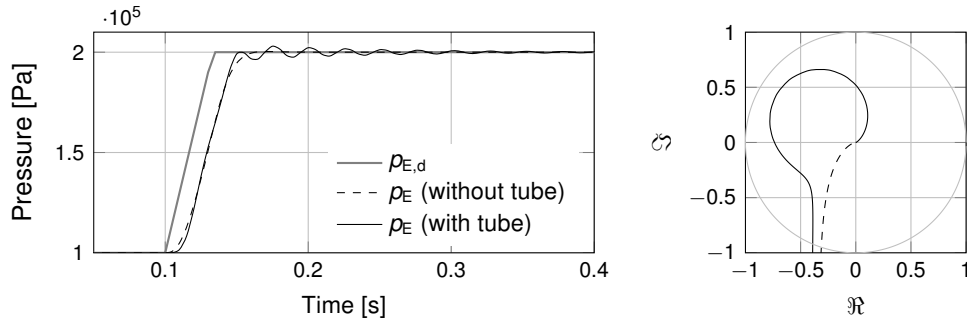


Figure 11. Time response and Nyquist diagram of the pressure control in a constant volume without considering the linear tube model in controller design.

with a damping factor of $D = 0.85$. According to the Nyquist diagram on the right, the closed loop has a large phase margin and is robust towards amplitude errors. The influence of the tube dynamics in the control loop can directly be seen in the initial lag at 0.1 s and the decaying oscillations around the desired pressure level. The Nyquist diagram reveals that the tube dynamics drive the control loop close to the stability limit.

Aside from considering the tube model in closed-loop analysis, it will be used for design, too. To increase robustness, the controller gain is lowered to meet certain criteria for phase and amplitude margin. The result is slower closed-loop dynamics, as can be seen in Figure 12 on the left. To improve the tracking behavior, the feedforward control [12]

$$P_{E,ff}(s) = \left(1 + \frac{k_1 s}{T_d s + 1} \right) P_{E,d}(s) \quad (25)$$

is added. The control law includes the desired pressure in the volume as well as its time derivative, that is attained by filtering and differentiation. The factor k_1 contains parameters of the plant including tube dynamics and the controller. The time response in Figure 12 now shows similar dynamics as in the nominal case. The oscillations are only slightly suppressed, but the robustness of the control loop is significantly higher (see Nyquist diagram in Figure 12 on the right). To avoid the pressure oscillations completely, the control law can

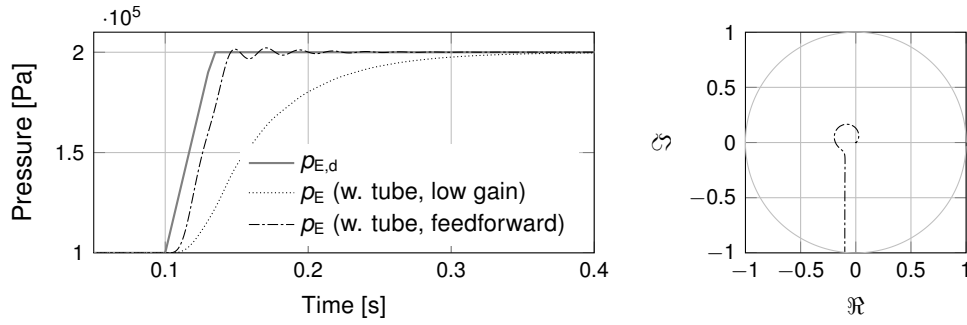


Figure 12. Time response and Nyquist diagram of the pressure control in a constant volume with consideration of the linear tube model in controller design.

be extended by a derivative term or replaced by a state space controller, since the simple tube model is also available for controller design and dimensioning. The presented model could then be applied as an observer to reconstruct the system's state variables.

5. CONCLUSION

The proposed pneumatic tube model is an approximation of the one-dimensional linear resistance compressible transmission line model of dynamic order two with special focus on the dominant eigenvalues. The augmentation with a pressure drop, based on the {C,b} method, accounts for essential non-linear friction effects in the tube. The model shows a good agreement with measurements in time as well as in frequency domain and has a lower order than comparable line models in literature.

The advantage of the model was shown for a pressure control application. By considering the tube dynamics in controller analysis and design, often neglected robustness issues were revealed and the control performance was increased.

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